(3) For higher mass numbers, the curve drops slowly and $\overline{BE}$ for uranium is about 7.6 MeV. Such nuclei are unstable and exhibit radioactive.

From Figure 9.24, if two light nuclei with A<28 combine with a nucleus with A<56, the binding energy per nucleon is more for final nucleus than initial nuclei. Thus, if the lighter elements combine to produce a nucleus of medium value A, a large amount of energy will be released. This is the basis of nuclear fusion and is the principle of the hydrogen bomb.

(4) If a nucleus of heavy element is split (fission) into two or more nuclei of medium value A, the energy released would again be large. The atom bomb is based on this principle and huge energy of atom bombs comes from this fission when it is uncontrolled. Fission is explained in the section 9.7

EXAMPLE 9.10

Compute the binding energy per nucleon of ${}_{2}^{4}He$ nucleus.

Solution

From Example 9.9, we found that the BE of ${}_{2}^{4}He$ =28.33 Mev

Binding energy per nucleon = $\overline{B.E}$ = 28.33 MeV/4 $\simeq$ 7 MeV.

9.5 NUCLEAR FORCE

Nucleus of the atoms contains protons and neutrons. From electrostatics, we learnt that like charges repel each other. In the nucleus, since the protons are separated by a distance of about a few fermi ($10^{-15}\,m$), they must exert on each other a very strong repulsive force. For example, the electrostatic repulsive force between two protons separated by a distance $10^{-15}\,m$

$$F = k \times \frac{q^2}{r^2} = 9 \times 10^9 \times \frac{\left(1.6 \times 10^{-19}\right)^2}{\left(10^{-15}\right)^2} \approx 230N$$

The acceleration experienced by a proton due to the force of 230 N is

$$a = \frac{F}{m} = \frac{230N}{1.67 \times 10^{-27}\,kg} \approx 1.4 \times 10^{29}\,m\,s^{-2}.$$

This is nearly 10^{28} times greater than the acceleration due to gravity. So if the protons in the nucleus experience only the electrostatic force, then the nucleus would fly apart in an instant. Then how the protons are held together in the nucleus?

From this observation, it was concluded that there must be a strong attractive force between protons to overcome the repulsive Coulombic force. This attractive force which holds the nucleons together is called strong nuclear force. The properties of the nuclear force were understood through various experiments carried out between 1930s and 1950s. A few properties of the nuclear force are

(i) The nuclear force is of very short range, acting only up to a distance of a few fermi. But inside the nucleus, the repulsive Coulomb force or attractive gravitational forces between two protons are much weaker than the nuclear force between two protons. Similarly, the gravitational force between two neutrons is also much weaker than nuclear force between the neutrons. So nuclear force is the strongest force in nature.

(ii) The nuclear force is attractive and acts with an equal strength between proton-proton, proton-neutron, and neutron – neutron.

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